The Well-Ordering Axiom is equivalent to comparability of cardinals

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/olp-set-theory-choice/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Card cut from the FREE slice of the lit pool for unit olp-set-theory-choice; the item labels equivwo and equivcompare are native to the statement span and need no navigation stubs. Statement span re-pinned to the enumerate body (713-717) because the gate span-start check requires the start within 2 lines of the item label at line 714; the enclosing thm opener (line 711, optional arg in ZF) and its closing end-of-environment line (line 718) are excluded to keep the carried span balanced, and the intro sentence (line 712) rides along as a context span.

Editorial scope. Single theorem card cut from the Open Logic Project unit olp-set-theory-choice (pinned-original-source.tex); statement and proof are the author's own text at source lines 713-717 (enumerate body of the thm environment, context line 712) and 720-736.

Allowed standard prerequisites (editorial).

- Ordinal representation theorem (set-theory-ordinals chapter, Theorem thmOrdinalRepresentation, source line 1020). cited at proof lines 723-725. Provides order-type isomorphisms from ordinals onto the well-ordered sets A and B.
- Proposition: ordinals are comparable by inclusion (set-theory-ordinals chapter, Proposition ordinalsaresubsets, source line 587). cited at proof line 726. Gives alpha a subset of beta or beta a subset of alpha for the two order-type ordinals.
- Hartogs' lemma (set-theory-choice chapter, Lemma HartogsLemma, source line 674). cited at proof lines 729-730. Supplies an ordinal beta with no injection into A, forcing the comparable injection to go from A into beta.

Source reference. Original theorem 0.26: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 1020–1020. Cited at proof lines 723-725: order-type isomorphisms for the two well-orderings.

Source reference. Original proposition 0.9: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 587–587. Cited at proof line 726: comparability of the two ordinals by inclusion.

Source reference. Original lemma 0.7: sources/olp-set-theory-choice/pinned-original-source.tex, source lines 674–674. Cited at proof lines 729-730: supplies the ordinal beta against which comparability is applied.

1. The Axiom of Well-Ordering
2. Either $A \preceq B$ or $B \preceq A$, for any sets A and B

The following claims are equivalent:

Proof. *1* $\Rightarrow$ *2* . Fix A and B. Invoking 1 , there are well-orderings $\langle A, R\rangle$ and $\langle B, S\rangle$. Invoking 0.26 , let $f\colon \alpha \to \langle A, R\rangle$ and $g\colon \beta \to \langle B, S\rangle$ be isomorphisms. By 0.9 , either $\alpha \subseteq \beta$ or $\beta \subseteq \alpha$. If $\alpha \subseteq \beta$, then $\circ f^{-1} g\colon A \to B$ is injection, and hence $A \preceq B$; similarly, if $\beta \subseteq \alpha$ then $B \preceq A$.

2 $\Rightarrow$ *1* . Fix A; by 0.7 there is some ordinal β such that $\beta \not\preceq A$. Invoking 2 , we have $A \preceq \beta$. So there is some injection $f\colon A \to \beta$, and we can use this injection to well-order the elements of A, by defining an order $\{\, \langle a, b\rangle \in A \times A \mid f(a) \in f(b) \,\}$. □